

SHRUTI SANJAY HALDE

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Summary

Software Engineer with an M.S. in Computer Science from USC and experience building scalable backend services, REST APIs, microservices, and cloud systems. Proficient in **Python, Java, Spring Boot, FastAPI, SQL, AWS, Docker, and Kubernetes**, with experience in distributed systems, database optimization, CI/CD, and production automation.

Education

University of Southern California

Master of Science in Computer Science; GPA: 3.52/4.00

Los Angeles, CA

Aug 2024 – May 2026

MCT's Rajiv Gandhi Institute of Technology

Bachelor of Engineering in Computer Engineering; CGPA: 9.76/10; University Rank 1

Mumbai, India

Aug 2019 – Jun 2023

Technical Skills

Languages: Python, Java, SQL, JavaScript, Bash, C, HTML/CSS

Frameworks & Backend: FastAPI, Flask, Django, Spring Boot, Spring MVC, React.js, REST APIs, Microservices, Maven

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Firebase

Cloud & DevOps: AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

Data & Analytics: Pandas, NumPy, ETL, Data Pipelines, Excel, Power BI, Scikit-learn, TensorFlow

Tools & Testing: Git, GitHub, Linux, Postman, Jira, PyTest

Core Concepts: OOP, Data Structures & Algorithms, System Design, Scalability, Distributed Systems, Performance Tuning, Authentication, RBAC, JSON, Async Processing

Experience

Hubcom Techno System LLP – A Talati & Partners Company

Software Engineer

Mumbai, India

May 2022 – Jul 2024

- Re-architected monolithic enterprise applications into **Java Spring Boot microservices**, improving service availability to **99.9%** and reducing production downtime by **40%** through fault isolation, health checks, and resilient service design.
- Designed and developed **REST APIs** using Java, Spring Boot, Python, FastAPI, and Flask processing up to **1M records/day**, improving data exchange efficiency by **46%** through pagination, payload optimization, and asynchronous processing.
- Implemented **Apache Kafka-based event-driven workflows** for asynchronous communication across distributed microservices, improving service decoupling, scalability, and reliability for real-time data processing.
- Improved **PostgreSQL and MongoDB** performance by **37%** through indexing, query optimization, execution-plan analysis, and schema refinements supporting high-volume production workloads.
- Containerized microservices using **Docker** and deployed **Kubernetes** workloads with rolling updates, health checks, and automated recovery to improve scalability, deployment consistency, and production reliability.
- Built automated testing and deployment workflows using **JUnit, Mockito, Postman, Jenkins CI/CD, and Git**, validating API behavior and accelerating release cadence from weekly to daily while maintaining production stability.
- Built **Python automation utilities** for batch validation, reporting, and operational workflows, reducing repetitive engineering tasks and improving production issue resolution through **Grafana monitoring and structured logging**.

Trent Ltd. – A TATA Enterprise

Data Analyst Intern

Mumbai, India

May 2025 – Jun 2025

- Developed an **Inventory Management dashboard** using **Power BI and SQL** to improve stock visibility, inventory tracking, and operational decision-making.
- Built automated **ETL and reporting workflows** using SQL Server, Python/Pandas, Power Automate, SharePoint, and Power BI Report Builder for data extraction, transformation, anomaly detection, and automated report delivery.

Projects

QueryBot – Natural Language to SQL | Python, T5, SQL-BART, Streamlit

Jan 2025 – May 2025

- Developed an **NLP-to-SQL semantic parsing system** using T5 and SQL-BART, fine-tuned on the Spider dataset to translate natural-language questions into executable SQL queries.
- Improved query-generation accuracy by **12% through schema-aware linking** and deployed an interactive **Streamlit** application supporting query execution and accuracy evaluation.

Weather Forecasting Application | Python, Flask, REST APIs, Cloud

Aug 2024 – Dec 2024

- Built a **Flask-based backend** integrating geolocation and weather APIs to collect, transform, and serve real-time location-based forecast data through REST endpoints and interactive visualizations.

Attendance System – Facial Recognition | Python, OpenCV, MySQL

Aug 2021 – May 2022

- Developed a real-time facial-recognition attendance system using **Python and OpenCV** with a **MySQL** backend for automated identity matching, attendance logging, and persistent record storage.

Publication & Certifications

Multiple Disease Prediction Using Machine Learning – IC-ICN 2023: Developed and evaluated Logistic Regression, SVM, and Random Forest models on healthcare datasets using preprocessing, feature selection, F1-score, and confusion matrices.

Certifications: Google Data Analytics Professional Certificate | Google Project Management Professional Certificate | DeepLearning.AI Mathematics for ML & Data Science | Elements of AI